Chapter 12. Sample Surveys

Sampling designs and stratified estimation

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12.1 Basic concepts of sample surveys

Motivation. A census is usually too slow, too costly, or even less accurate than a well-designed sample. To use a sample well we must define precisely which population is studied and which errors can arise.

12.1.1 What is a sample survey

To learn about a population we can survey all of it or only part of it and estimate the whole from that part.

Definition (Census and sample survey)

(a) Surveying the entire population is a **census** (complete enumeration).
(b) Drawing a sample from part of the population and inferring population characteristics from it is a **sample survey**.

In theory a census gives exact information. In practice it is often impossible, or it introduces more errors. Surveying the attitudes of every Korean teenager towards the opposite sex, for instance, would take years for one person. The results could be outdated before they appeared, and hiring tens of thousands of interviewers would be very costly. For opinion and attitude surveys a sample survey is the sensible choice.

Definition (Populations, frames and units)

(a) The **target population** is the population we actually want to study. It must be defined precisely. In a survey of "university students", for example, we must decide whether students on leave or graduate students are included.
(b) The **sampling frame** (or simply **frame**) is the list or rule that identifies the units of the target population available for sampling. It is the accessible subset from which the sample is actually drawn.
(c) The **sampled population** is the population actually covered by the frame. If the target is "all adults in Korea" but the frame is "adults listed in the telephone directory", unlisted adults are excluded, so the sampled and target populations differ.
(d) A **sampling unit** is an element of the population that can be selected. In a survey of students' attitudes it is each student; in a survey of farm household incomes it is each household.
(e) The **sampling design** specifies which units are selected and how. It is chosen to suit the parameters to be estimated and the structure of the population, so that estimation is as efficient as possible.

12.1.2 Errors and bias

Definition (Sampling and nonsampling error)

(a) **Sampling error** is the statistical error that arises because we infer about the whole population from part of it.
(b) **Nonsampling error** is every other error: the gap between target and sampled populations, interviewer mistakes, poorly worded questionnaires, and so on.

A sample survey contains both kinds of error. Sampling error is usually designed to shrink at the rate $1/\sqrt{n}$. Nonsampling error does not depend on n and can even be larger in a census. Hence

the total error first decreases as n grows, but beyond some point it no longer decreases, because nonsampling error dominates.

The size of the sampling error can be worked out theoretically and controlled through n. Nonsampling error is hard to estimate or predict, so it must be reduced by care in planning, data collection and analysis. Its main sources are the following.

(i) **Faulty questions**: unclear or leading questions distort answers.

(ii) **Wrong answers and interviewer inexperience**: respondents may answer incorrectly or misunderstand the question (**response error**), or the interviewer may convey the purpose wrongly or be careless.

(iii) **Nonresponse error** (nonresponse bias): selected people do not respond. A low response rate makes the sample unrepresentative, especially when nonrespondents have particular tendencies.

(iv) **Coverage error**: some units are missing from (or wrongly included in) the frame, or the frame differs greatly from the target population.

(v) **Processing error**: errors in recording or in computer entry.

For an estimator $\hat{\theta}$ of θ, the difference $E(\hat{\theta}) - \theta$ is its bias. In survey practice the word **bias** is used more broadly: it also covers the systematic distortions caused by the nonsampling errors above. The most serious source of bias is a flawed sampling design.

Case study: the 2015 UK general election polls

The UK Parliament has an elected House of Commons and an appointed or hereditary House of Lords. Real power lies with the Commons, whose majority leader becomes prime minister. Since 1992 the Conservatives had never won a Commons majority. Before the 2015 election, polls taken over five weeks of campaigning predicted that neither party would win a majority and that Labour and the Conservatives would be roughly level. In fact the Conservatives won 331 seats, 91 more than Labour, a clear majority. The British Polling Council's inquiry identified two main causes.
(i) *Wrongly chosen sampled population.* The frames were unrepresentative. Labour supporters were over-represented and Conservative supporters under-represented. Respondents were mostly urban workers who answer surveys easily, while older Conservative voters were hard to reach.
(ii) *Neglect of nonresponse.* Respondents differed from nonrespondents. Conservative supporters likely had a lower response rate, so their real support was underestimated.

Similar errors are common. Examples include surveying national attitudes using only parents at schools in particular apartment complexes, or telephone polls that sample only landline users and exclude mobile users. Such frames produce serious bias, which shows how important the sampling design is.

12.2 Sampling methods

Motivation. Different designs trade off cost, convenience and precision. The basic designs (simple random, systematic, stratified, cluster and stratified cluster sampling) are the building blocks of real surveys.

12.2.1 Simple random sampling

Definition (Simple random sampling)

Drawing n units from a population of N units so that every unit has the same chance of selection, i.e. so that all $\binom{N}{n}$ possible samples are equally likely, is **simple random sampling** (SRS). In practice:
(i) number the units 1 to N;
(ii) select n of the numbers at random.

A random number generator is normally used. For a small population we can also draw lots from shuffled slips of paper.

Theorem (Sample mean under SRS without replacement)

Let a finite population have values $x_1, \dots, x_N$ with mean $\mu = \frac{1}{N}\sum_k x_k$ and variance $\sigma^2 = \frac{1}{N}\sum_k (x_k - \mu)^2$. For an SRS of size n without replacement with sample mean $\overline{X}$ and sample variance S^2,

$$E(\overline{X}) = \mu, \qquad \operatorname{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}, \qquad E(S^2) = \frac{N}{N-1}\sigma^2.$$

Consequently $\left(1 - \frac{n}{N}\right)\frac{S^2}{n}$ is an unbiased estimator of $\operatorname{Var}(\overline{X})$.

Proof. [Supplement]

Step 1. Let $\delta_k = 1$ if unit k is in the sample and $\delta_k = 0$ otherwise, so that $\overline{X} = \frac{1}{n}\sum_{k=1}^{N} x_k \delta_k$. By symmetry of SRS, for $k \neq l$,

$$P(\delta_k = 1) = \frac{n}{N}, \qquad P(\delta_k = \delta_l = 1) = \frac{n(n-1)}{N(N-1)}.$$

Hence $E(\overline{X}) = \frac{1}{n}\sum_k x_k \frac{n}{N} = \mu$.

Step 2. From Step 1,

$$\operatorname{Var}(\delta_k) = \frac{n}{N}\left(1 - \frac{n}{N}\right) = \frac{n(N-n)}{N^2}, \qquad \operatorname{Cov}(\delta_k, \delta_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3. Since $\sum_k \delta_k = n$ is constant, $\sum_k x_k\delta_k$ and $\sum_k d_k\delta_k$ (with $d_k = x_k - \mu$) have the same variance. Using $\sum_k d_k = 0$, so that $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2 = -N\sigma^2$,

$$\operatorname{Var}\Big(\sum_k d_k\delta_k\Big) = \frac{n(N-n)}{N^2}\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big] = \frac{n(N-n)}{N^2}\, N\sigma^2\Big(1 + \frac{1}{N-1}\Big)$$
$$= \frac{n(N-n)}{N-1}\,\sigma^2.$$

Dividing by n^2 gives $\operatorname{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}$.

Step 4. Let $X_1, \dots, X_n$ be the sample values. Each X_i has mean μ and variance σ^2, and

$$E\sum_{i=1}^{n}(X_i - \overline{X})^2 = E\sum_{i=1}^{n}(X_i - \mu)^2 - n\,E(\overline{X} - \mu)^2 = n\sigma^2 - \frac{N-n}{N-1}\sigma^2 = \frac{N(n-1)}{N-1}\sigma^2.$$

So $E(S^2) = \frac{N}{N-1}\sigma^2$. Finally,

$$E\Big[\Big(1 - \frac{n}{N}\Big)\frac{S^2}{n}\Big] = \frac{N-n}{N}\cdot\frac{N}{N-1}\cdot\frac{\sigma^2}{n} = \operatorname{Var}(\overline{X}).$$

■

12.2.2 Systematic sampling

Definition (Systematic sampling)

In **systematic sampling** the units are listed in order and selected at a fixed interval. With population size N and sample size n, the **sampling interval** is $k = N/n$. One unit among the first k is chosen at random, and then every k-th unit after it is taken.

For example, to draw $n = 200$ from $N = 10000$ use $k = 10000/200 = 50$. If the random start is 23, the sample consists of units $23, 73, 123, 173, 223, 273, \ldots, 9973$. The method is simpler than SRS and spreads the sample evenly over the population.

Exit polls, which ask voters leaving a polling station how they voted, use systematic sampling. The frame size N and the sample size n are hard to know in advance, so the interval k is fixed beforehand. The joint exit polls of the three Korean broadcasters for the 20th presidential election (2022) and the 22nd National Assembly election (2024) used $k = 5$.

Systematic sampling is convenient and is common in large surveys. If the list has a periodic pattern whose period matches k, however, it can produce serious bias.

12.2.3 Stratified random sampling

Definition (Stratified random sampling)

Divide the population into several groups, the **strata**, draw an SRS within each stratum, and use all of these together as the sample. This is **stratified random sampling** (or simply **stratified sampling**).

Strata are formed so that units are homogeneous within a stratum and heterogeneous across strata. Then every kind of unit appears in the sample, which raises the precision of the estimates. For this reason stratification is regarded as the theoretically most important sampling method.

Suppose, for example, that students' attitudes differ by college and by year. College and year are then natural stratification variables. If students of the same college and year think alike, a small sample from each stratum describes it accurately, and precision for the whole university improves greatly. Strata should therefore be chosen so that units within each are as homogeneous as possible.

12.2.4 Cluster sampling

Definition (Cluster sampling)

In **cluster sampling** the population is divided into subpopulations called **clusters**. Some clusters are chosen by SRS. Either every unit in a chosen cluster is surveyed (**single-stage cluster sampling**) or an SRS of units is drawn within each chosen cluster (**two-stage cluster sampling**).

Because whole clusters (or samples from them) are surveyed, cluster sampling works well when each cluster resembles the population, i.e. when clusters differ little from one another and each represents the whole. Its main advantage is lower cost. If students' attitudes depend on year but not on major, for example, we can take departments as clusters, choose a few departments at random, and survey all or some of their students. This is much cheaper than SRS or stratified sampling over the whole university.

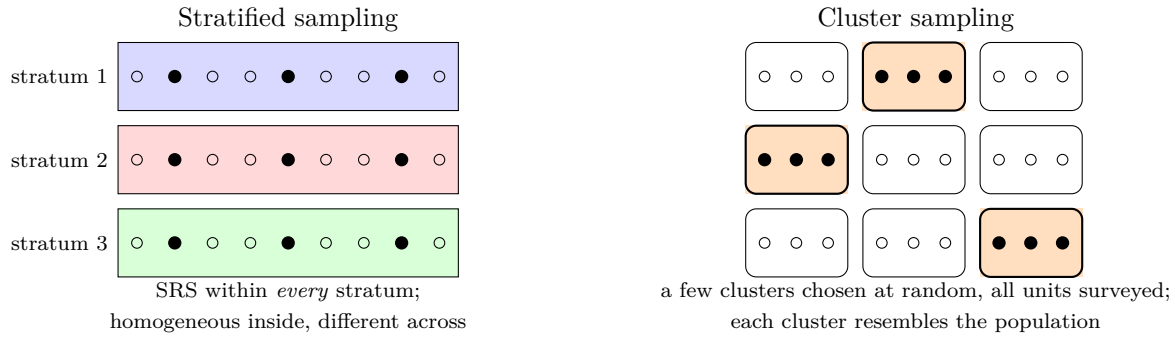


Figure 12.1: Stratified sampling versus (single-stage) cluster sampling; filled dots are sampled units.

12.2.5 Stratified cluster sampling

Definition (Stratified cluster sampling)

In **stratified cluster sampling** the population is first stratified into primary units. Within each stratum, clusters are drawn as primary sampling units, and the chosen clusters are surveyed completely or sampled further. Surveying the chosen clusters completely is **stratified single-stage cluster sampling**. Drawing secondary units from them by SRS is **stratified two-stage cluster sampling**.

Real surveys combine the designs above. Suppose students' attitudes differ considerably between colleges, but little between departments within a college. Take colleges as strata and departments as clusters (primary units). Choose departments at random within each college, and then draw students (secondary units) from the chosen departments by SRS. This is stratified two-stage cluster sampling.

12.3 Estimation under stratified random sampling

Motivation. Stratified sampling is the most important design. Here we derive its estimators of a population mean and a population proportion, with their standard errors and confidence intervals.

A population of N units is divided into h strata of sizes N_i ($\sum_i N_i = N$). The most common way to split the total sample size n among the strata is **proportional allocation**, where each stratum's sample size is proportional to its size:

$$n_i = n\Big(\frac{N_i}{N}\Big) \qquad (i = 1, 2, \dots, h).$$

For means, let μ_i and σ_i^2 be the mean and variance of stratum i, with $\sigma_i^2 = \frac{1}{N_i}\sum(x - \mu_i)^2$ over the stratum. For proportions, let P_i be the proportion of units in stratum i with the attribute.

Definition 12-1 (Estimation of the population mean)

(i) Estimator:

$$\overline{X}_{\text{st}} = \frac{N_1}{N}\overline{X}_1 + \frac{N_2}{N}\overline{X}_2 + \cdots + \frac{N_h}{N}\overline{X}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \overline{X}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{\sigma_i^2}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{S_i^2}{n_i}}\,.$$

Here $\overline{X}_i$ and S_i^2 are the sample mean and sample variance of stratum i, and the subscript st stands for "stratified".

Justification. [Supplement] The samples from different strata are independent SRSs without replacement.

Step 1. By the theorem of Section 12.2.1, $E(\overline{X}_i) = \mu_i$. Hence $E(\overline{X}_{\mathrm{st}}) = \frac{1}{N}\sum_i N_i\mu_i = \mu$, so $\overline{X}_{\mathrm{st}}$ is unbiased.

Step 2. By independence across strata and the same theorem,

$$\mathrm{Var}(\overline{X}_{\mathrm{st}}) = \frac{1}{N^2}\sum_i N_i^2\,\mathrm{Var}(\overline{X}_i) = \frac{1}{N^2}\sum_i N_i^2\,\frac{N_i-n_i}{N_i-1}\cdot\frac{\sigma_i^2}{n_i},$$

which is (ii).

Step 3. The same theorem shows that $(1-n_i/N_i)S_i^2/n_i$ is unbiased for $\mathrm{Var}(\overline{X}_i)$. Summing these estimates with weights N_i^2/N^2 gives an unbiased estimator of $\mathrm{Var}(\overline{X}_{\mathrm{st}})$, and its square root is (iii). ■

Definition 12-2 (Estimation of the population proportion)

(i) Estimator:

$$\hat{P}_{\mathrm{st}} = \frac{N_1}{N}\hat{P}_1 + \frac{N_2}{N}\hat{P}_2 + \cdots + \frac{N_h}{N}\hat{P}_h = \frac{1}{N}\sum_{i=1}^{h} N_i\hat{P}_i.$$

(ii) Variance of the estimator:

$$\mathrm{Var}(\hat{P}_{\mathrm{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(\frac{N_i-n_i}{N_i-1}\Big)\frac{P_i(1-P_i)}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i}}\,.$$

Here $\hat{P}_i = X_i/n_i$ is the sample proportion of stratum i, where X_i is the number of sampled units in stratum i with the attribute.

Justification. [Supplement] Code each unit as 1 (has the attribute) or 0. Then $\mu_i = P_i$, $\sigma_i^2 = P_i(1-P_i)$ and $\overline{X}_i = \hat{P}_i$, so (i) and (ii) are the special cases of Definition 12-1.

For the sample variance of 0–1 data, $S_i^2 = \frac{n_i}{n_i-1}\hat{P}_i(1-\hat{P}_i)$. The exactly unbiased variance estimate is therefore

$$\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i-1}.$$

Formula (iii) with n_i in the denominator differs from it by the factor $(n_i-1)/n_i \approx 1$. Example 12-2 uses the exact version with $n_i - 1$. ■

Finite population correction and confidence intervals

In part (ii) of Definitions 12-1 and 12-2, the factor $\dfrac{N_i - n_i}{N_i - 1}$ appears because the population is finite. It is the **finite population correction factor**. In the estimated standard error it is replaced by $\dfrac{N_i - n_i}{N_i} = 1 - \dfrac{n_i}{N_i}$, which makes the variance estimate unbiased (see the theorem of Section 12.2.1). Part (iii) of each definition replaces σ_i^2 and P_i by their estimates S_i^2 and $\hat{P}_i$.

The 95% confidence intervals for the mean and the proportion are

$$\bar{x}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\overline{X}_{\text{st}}) = \bar{x}_{\text{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1 - \frac{n_i}{N_i}\Big)\frac{s_i^2}{n_i}},$$

$$\hat{p}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}}) = \hat{p}_{\text{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1 - \frac{n_i}{N_i}\Big)\frac{\hat{p}_i(1-\hat{p}_i)}{n_i}}.$$

The half-widths $z_{0.025}\widehat{\text{SE}}(\overline{X}_{\text{st}})$ and $z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}})$ are called the **margin of error**, also known as the error limit or allowable error.

Example 12-1 (Household living expenses)

To survey monthly living expenses per household in a small city, the city was divided by standard of living into 3 regions (strata), and a stratified sample gave the following data (unit: 100,000 won).

Stratum	N_i	n_i	$\bar{x}_i$	s_i^2
1	2,800	30	18.5	12.2
2	5,600	60	15.2	11.3
3	3,400	30	10.7	10.5
Total	$N = 11{,}800$	$n = 120$		

Estimate the mean monthly living expenses per household and give a 95% confidence interval.

Solution. The estimate is

$$\bar{x}_{\text{st}} = \frac{1}{11800}(2800 \times 18.5 + 5600 \times 15.2 + 3400 \times 10.7) = 14.69.$$

The estimated variance is

$$\widehat{\text{Var}}(\overline{X}_{\text{st}}) = \frac{1}{11800^2}\Big\{2800^2\Big(1 - \frac{30}{2800}\Big)\Big(\frac{12.2}{30}\Big) + 5600^2\Big(1 - \frac{60}{5600}\Big)\Big(\frac{11.3}{60}\Big) + 3400^2\Big(1 - \frac{30}{3400}\Big)\Big(\frac{10.5}{30}\Big)\Big\} = 0.09342.$$

So $\widehat{\text{SE}}(\overline{X}_{\text{st}}) = \sqrt{0.09342} = 0.3056$, and

$$\bar{x}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\overline{X}_{\text{st}}) = 14.69 \pm 1.96 \times 0.3056 = 14.69 \pm 0.60.$$

The 95% confidence interval is $(14.09,\ 15.29)$, i.e. about 1.41 to 1.53 million won.

Example 12-2 (Support for a university policy)

A university grouped similar colleges into 3 strata A, B, C and surveyed students on leave about a university policy by stratified sampling.

College (stratum)	N_i	n_i	$\hat{p}_i$
A	3,500	35	0.56
B	4,500	45	0.60
C	2,000	20	0.40

Estimate the support rate and give a 95% confidence interval.
Solution. The estimate is

$$\hat{p}_{\text{st}} = \frac{1}{10000}(3500 \times 0.56 + 4500 \times 0.60 + 2000 \times 0.40) = 0.546.$$

Using $n_i - 1$ in the denominators,

$$\widehat{\text{Var}}(\hat{P}_{\text{st}}) = \frac{1}{10000^2}\Big\{3500^2\Big(1 - \frac{35}{3500}\Big)\Big(\frac{0.56 \times 0.44}{35-1}\Big) + 4500^2\Big(1 - \frac{45}{4500}\Big)\Big(\frac{0.60 \times 0.40}{45-1}\Big) + 2000^2\Big(1 - \frac{20}{2000}\Big)\Big(\frac{0.40 \times 0.60}{20-1}\Big)\Big\} = 0.00247.$$

So $\widehat{\text{SE}}(\hat{P}_{\text{st}}) = \sqrt{0.00247} = 0.0497$, and

$$\hat{p}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}}) = 0.546 \pm 1.96 \times 0.0497 = 0.546 \pm 0.097.$$

The 95% confidence interval is (0.449, 0.643), i.e. 44.9% to 64.3%. (The book centres the interval at 0.564 by a transposition slip and reports (0.467, 0.661); the margin of error 0.097 is the same.)

12.4 Problems in practical surveys

Motivation. A published "margin of error" covers only sampling error. Misunderstanding randomness and writing poor questionnaires create nonsampling errors that no formula reports.

Media reports of opinion polls often contain statements like this one. *The survey used automated wireless calls to virtual (safe) mobile numbers, with a sample of 1,003, a response rate of* 2.3%, *and a sampling error of* ±3.1 *percentage points at the* 95% *confidence level.* The "sampling error" 3.1% is the margin of error of Section 12.3. It describes only the error from surveying part of the population instead of all of it, i.e. the size of the sampling error. Nonsampling error is not included. Professional survey firms design their samples on sound theory and experience, so their nonsampling errors are comparatively small, and the stated margin of error is then a good indicator of reliability. When non-experts run surveys, however, nonsampling error can grow sharply. Two typical sources are misunderstanding or ignoring randomness, and poorly worded questions.

12.4.1 The concept of randomness

Drawing a sample at random does not mean drawing it any old way. In random sampling, units are selected with probabilities fixed in advance. As noted in Section 12.1.2, if the sample is not drawn from the whole population according to statistical theory, bias and nonsampling error can become severe.

Suppose a graduate student studies "generational differences in attitudes to the division of household chores and gender equality". A sample drawn with all Korean households as the sampled population

would be beyond one person's time and budget, so shortcuts get used. For instance, questionnaires might be handed to high-school students in one apartment complex and filled in by their parents. The sampled population is then restricted to families in that complex with a high-school child. Childless families and the elderly are almost excluded, so a "generational" analysis becomes difficult. Moreover, households in one complex share social and economic characteristics, so the sampled population differs greatly from the target population.

The 2015 UK polls of Section 12.1 failed in the same way. Such errors come from reading "random" as "haphazard", or from choosing convenient methods to save cost or effort. Random sampling means fixing each unit's selection probability in advance and then selecting by chance according to those probabilities. Ignoring this can make nonsampling error and bias severe.

12.4.2 Writing the questionnaire

Questions must neither lead respondents to a particular answer nor confuse them with vague wording.

(a) *Leading questions.* "Do you think ancestral rites should be kept in the traditional way, to honour the ancestors and teach tradition to our children?" and "Do you think ancestral rites should be simplified to fit social development and changing values?" both invite a "yes". If asked separately, the proportion favouring keeping rites and the proportion favouring simplifying them may well sum to more than 100%.

(b) *Ambiguous questions.* "Are you satisfied with your current life?" intends to measure life satisfaction, but "current life" is undefined, so respondents may be confused. To measure satisfaction properly, ask about several aspects (work, marriage, family, leisure, friendships) and combine them. "Do you think you have many friends?" depends on how "many" is read: one person regards 3 friends as many, another regards 10 as few.

(c) *Campaigning disguised as a poll.* Around elections, questions such as "Do you know that candidate X devoted his life to the democracy movement?" or "Do you know that candidate X is an economic expert and former minister?" clearly aim to promote a candidate. Such use of a poll abuses statistical surveys.

(d) *Pretesting.* Problems are best found and fixed in a **pilot survey**. Consider "What is your monthly leisure budget? (1) under 100,000 won (2) 100–500 thousand (3) 500 thousand–1 million (4) over 1 million won". Given to university students, most will choose (1). The answers then show only that student leisure spending is low and say nothing useful about its distribution. A pilot survey reveals this easily, and the categories can be revised.

(e) *Length.* With too many questions, respondents tire and answer carelessly. Limit the number of items, and offering fair compensation for the respondent's time also helps.

Every item must be reviewed carefully. However sophisticated the sampling design, a badly written questionnaire yields distorted information.